4.7.4 Case study 4: X-47B – Autonomous naval combat operations (2012–2015)

The Northrop Grumman X-47B became the first UAV to autonomously operate from an aircraft carrier, achieving catapult-assisted take-off and arrested recovery aboard the USS George H.W. Bush (CVN-77) in 2013. Developed under the US Navy's UCAS-D program, it demonstrated the integration of autonomy, precision navigation, and stealth in maritime operations. In subsequent tests, the X-47B performed aerial refueling and coordinated airspace sharing with manned aircraft, validating key technologies for carrier-based UCAVs. Its success directly influenced the design of the MQ-25 Stingray, the US Navy's first operational aerial refueling UAV. The X-47B's demonstration marked a paradigm shift in naval aviation, proving that unmanned systems could safely and effectively integrate into carrier operations, thereby expanding the reach and endurance of future naval air wings.

4.7.5 Case study 5: Wing Loong I – Exported combat efficiency and global reach (2015–Present)

The Wing Loong I, developed by AVIC China, emerged as a cost-effective, export-oriented UAV that expanded unmanned aerial capabilities to numerous developing nations. It combined ISR and strike capabilities, carrying precision-guided munitions and advanced EO/IR sensors. Used extensively by Saudi Arabia, Egypt, and the UAE, the Wing Loong I has participated in counterinsurgency and border surveillance operations, particularly in the Middle East and Africa. Its affordability, reliability, and modular design allowed nations with limited defense budgets to acquire advanced UAV capabilities. This widespread adoption demonstrated China's success in democratizing UAV technology, reshaping global defense markets, and challenging Western dominance in the drone industry.